Delta Hedging

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Questions and scope

A question

Is there a hedging strategy that has the smallest next-day error?

An approach

We will analyse this question based on 5 models of hedging.

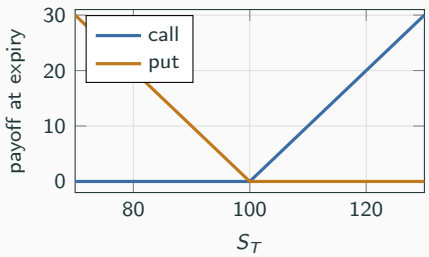
Calls and puts: an example

Synthetic example. $S_0 = K = 100$, $\tau = 0.25$, $r = 3\%$, $q = 1.3\%$. With $(v_0, \theta, \kappa, \xi, \rho) = (0.04, 0.04, 2, 0.6, -0.7)$, Heston gives a call price of \$3.923 and a Black–Scholes implied volatility of 18.70%.

Call: right to buy at K; payoff $\max(S_T - K, 0)$, initial price \$3.92.

Put: right to sell at K; payoff $\max(K - S_T, 0)$, initial price \$3.50.

S_T	80	100	120
Call payoff	0	0	20
Call profit	−3.92	−3.92	+16.08
Put payoff	20	0	0
Put profit	+16.50	−3.50	−3.50



What is hedging?

A short call is exposed to changes in the stock price. Buying shares creates an opposing stock exposure: a rise benefits the shares while hurting the short call, and a fall has the reverse effect.

This offset is local to small stock moves. When a stock move also changes variance, the call responds to both shocks while the shares respond only to the stock price. Passage of time changes the call's value as well.

A share hedge reduces directional exposure but does not remove volatility or time risk.

Four models for the hedge ratio h

	Model	Hedge ratio h	Example	Volatility
1	Black–Scholes	$\Delta_{BS} = e^{-q\tau}\Phi(d_1)$	0.535	constant
2	Heston	$\partial_S C$	0.628	random (v held fixed)
3	Heston min.-variance	$\partial_S C + \frac{\rho\xi}{S}\,\partial_v C$	0.461	random, corr. with S
4	Hull–White	$\Delta_{BS} + \frac{\mathcal{V}_{BS}}{S\sqrt{\tau}}\left(a + bd + cd^2\right)$	0.464	learned from data

Here d is the call-equivalent Black–Scholes delta (also for puts). The regression coefficients use 100 simulated paths; the worked comparison then reprices one call under two prescribed shocks. A second call can also hedge variance exposure.

Model 1: Black–Scholes

Model 1: Black–Scholes PDE and price

Assumption. Under the physical measure, $dS_t = (\mu - q)S_t\,dt + \sigma S_t\,dW_t$ with constant σ. Pricing replaces μ by the risk-free rate r.

PDE.

$$\partial_t V + \tfrac{1}{2}\sigma^2 S^2 \partial_{SS} V + (r-q)S\partial_S V - rV = 0, \qquad V(S,T) = \text{payoff}.$$

Call price.

$$C = Se^{-q\tau}\Phi(d_1) - Ke^{-r\tau}\Phi(d_2), \qquad d_{1,2} = \frac{\ln(S/K) + (r - q \pm \tfrac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}.$$

The put follows from put–call parity: $P = C - Se^{-q\tau} + Ke^{-r\tau}$.

Model 1: Black–Scholes delta

Differentiating the option price with respect to spot gives

$$\Delta_{BS} = \partial_S V = \begin{cases} e^{-q\tau}\Phi(d_1) & \text{call}, \\ e^{-q\tau}[\Phi(d_1) - 1] & \text{put}. \end{cases}$$

For the synthetic at-the-money call with $\tau = 0.25$, $r = 3\%$, $q = 1.3\%$ and $\sigma_{\text{imp}} = 18.70\%$,

$$d_1 = 0.0922, \qquad \Delta_{BS} = e^{-0.013(0.25)}\Phi(0.0922) = \mathbf{0.535}.$$

A model-generated volatility skew

For these model-generated prices, implied volatility σ_{imp} solves $C_{BS}(S, K, \tau, r, q, \sigma_{\text{imp}}) = C_{\text{Heston}}(S, K, \tau)$.

Strike K	80	85	90	95	100	105	110	115	120
Implied vol (%)	27.3	25.3	23.2	21.0	18.7	16.5	15.0	14.5	14.6

three-month options with $S = 100$; the $K = 100$ call is our example

- Under Black–Scholes this row would be flat.
- Δ_{BS} at an option's own implied vol assumes that implied vol *does not move* when S moves ("sticky strike").
- Here $\rho = -0.7$, so stock and variance shocks are negatively correlated.

Model 2: Heston

Model 2: Heston (1993)

SDEs.

$$dS_t = (\mu - q)S_t\,dt + \sqrt{v_t}\,S_t\,dW_t^S$$

$$dv_t = \kappa(\theta - v_t)\,dt + \xi\sqrt{v_t}\,dW_t^v$$

$$\mathrm{Cov}\left(dW_t^S,\, dW_t^v\right) = \rho\,dt$$

v_t	variance at time t
θ	long-run variance
κ	speed of mean reversion
ξ	volatility of variance (vol of vol)
ρ	correlation of price and variance

Example: $v_0 = \theta = 0.04$, $\kappa = 2$, $\xi = 0.6$, $\rho = -0.7$.

Feller condition

$2\kappa\theta \geq \xi^2$ keeps v_t away from zero; our illustrative parameters violate it ($0.16 < 0.36$).

Pricing: Lewis' formula

The price today of a call with time τ to expiry is

$$C_\tau = S_0 e^{-q\tau} - \frac{\sqrt{S_0 K}\, e^{-(r+q)\tau/2}}{\pi} \int_0^\infty \frac{\mathrm{Re}\left[e^{iuk}\, \varphi(u - \frac{i}{2})\right]}{u^2 + \frac{1}{4}}\, du,$$

where $k = \ln(S_0/K) + (r-q)\tau$ and $\varphi(u) := \mathbb{E}\left[e^{iuX_\tau}\right]$ with $X_\tau := \ln(S_\tau/F_\tau)$ and $F_\tau = S_0 e^{(r-q)\tau}$ being the forward price.

In our example: $C_\tau = \$3.923$ for the three-month ATM call.

Calibration on a synthetic option surface

Choose $(v_0, \kappa, \theta, \xi, \rho)$ that minimizes

$$\sum_i \left(\frac{C_i^{\text{Heston}} - C_i^{\text{market}}}{\mathcal{V}_{BS,i}} \right)^2 \quad \text{over} \quad (v_0, \kappa, \theta, \xi, \rho).$$

The vega scaling approximates an implied-volatility least squares criterion.

Parameter	True	Mean fit	Standard deviation
v_0	0.0400	0.0397	0.0006
κ	2.0000	1.9797	0.1693
θ	0.0400	0.0410	0.0012
ξ	0.6000	0.6117	0.0262
ρ	−0.7000	−0.6959	0.0064

Model 2: the Heston delta

Differentiation gives the spot and instantaneous-variance sensitivities:

$$\partial_S C = e^{-q\tau} - \frac{c}{S}\int_0^\infty \mathrm{Re}\left[\left(\tfrac{1}{2} + iu\right) f(u)\right] du, \qquad \partial_v C = -\,c\int_0^\infty \mathrm{Re}\left[B\left(u - \tfrac{i}{2}\right) f(u)\right] du,$$

where $c := \sqrt{SK}\, e^{-(r+q)\tau/2}/\pi$, $f(u) := \dfrac{e^{iuk}\varphi(u - \frac{i}{2})}{u^2 + \frac{1}{4}}$.

- A long call hedged with the plain Heston delta holds $-\partial_S C$ shares.
- In Heston, dv and dS are correlated: when S moves, v moves too.
- ATM three-month call: $\partial_S C = \mathbf{0.628}$ vs. $\Delta_{BS} = 0.535$; $\partial_v C = 39.94$.

Model 3: Heston minimum-variance delta

Model 3: the minimum-variance delta

A Heston option has two exposures: $dC \approx \partial_S C \, dS + \partial_v C \, dv$.

Upon minimizing $\mathrm{Var}(dC - h\, dS)$:

$$h^* = \frac{\mathrm{Cov}(dC, dS)}{\mathrm{Var}(dS)} = \partial_S C + \partial_v C \, \frac{\mathrm{Cov}(dv, dS)}{\mathrm{Var}(dS)} = \partial_S C + \partial_v C \, \frac{\rho \xi v S \, dt}{v S^2 \, dt}.$$

Minimum-variance delta

$$\Delta_{MV} = \partial_S C + \frac{\rho \, \xi}{S} \, \partial_v C$$

ATM 3 month call example:

$$\Delta_{MV} = 0.628 + \frac{-0.7 \times 0.6}{100} \times 39.94 = 0.628 - 0.168 = \mathbf{0.461}.$$

Synthetic example: one day, one $1 move

Three-month ATM call; $v_0 = \theta = 0.04$, $\kappa = 2$, $\xi = 0.6$, $\rho = -0.7$. Price \$3.923, implied vol 18.70%.

Hedge	Shares	Error ($S \uparrow$)	Error ($S \downarrow$)
Black–Scholes Δ_{BS}	0.535	-0.084	$+0.064$
Plain Heston $\partial_S C$	0.628	-0.177	$+0.158$
Minimum-variance Δ_{MV}	0.461	$\mathbf{-0.010}$	$\mathbf{-0.010}$

Model 4: Hull–White

Model 4: Hull & White (2017)

Keep Black–Scholes, and learn the vol–spot relation from data:

$$\Delta_{HW} = \Delta_{BS} + \frac{\mathcal{V}_{BS}}{S\sqrt{\tau}}\big(a + b\,d + c\,d^2\big),$$

where (a, b, c) minimizes the following function:

$$Q(a,b,c) := \sum_{(i,t)\,\in\,\text{training set}} \Big[e_{BS} - \frac{\mathcal{V}_{BS}}{S_t\sqrt{\tau}}\,(S_{t+1} - S_t)\big(a + bd + cd^2\big)\Big]^2$$

where $d = \Delta_{BS}$ for calls and $d = \Delta_{BS} + e^{-q\tau}$ for puts; and
$e_{BS} = (C_{t+1} - C_t) - \Delta_{BS}(S_{t+1} - S_t)$.

Fitted on simulated data: $(a, b, c) = (-0.305,\ 0.386,\ -0.279)$. For the call above,

$$\Delta_{HW} = 0.535 + 0.396 \times (-0.178) = \mathbf{0.464} \quad \text{vs.} \quad \Delta_{MV} = \mathbf{0.461}.$$

Extension: a second option

Delta–vega hedging

A second option H, the one-year at-the-money call, supplies a volatility-sensitive hedge. The position is long the three-month call, short n copies of H, and short h shares.

$$n_{BS} = \frac{\mathcal{V}_{BS}(C)}{\mathcal{V}_{BS}(H)} = 0.511, \qquad h_{BS} = \Delta_{BS}(C) - n_{BS}\Delta_{BS}(H) = 0.246.$$

For Heston,

$$n_H = \frac{\partial_v C}{\partial_v H} = 1.014, \qquad h_H = \partial_S C - n_H \partial_S H = -0.050.$$

Synthetic example: all hedges on one day

Three-month ATM call; $\delta S = \pm\$1$ and the correlated variance shock is $\delta v = (\rho\xi/S)\delta S = \mp 0.0042$. Both options have one trading day less to expiry.

Hedge	Shares h	One-year calls n	Error ($S\uparrow$)	Error ($S\downarrow$)
Unhedged	0	—	$+0.451$	-0.471
1 Black–Scholes Δ_{BS}	0.535	—	-0.084	$+0.064$
2 Heston $\partial_S C$	0.628	—	-0.177	$+0.158$
3 Heston Δ_{MV}	0.461	—	$\mathbf{-0.010}$	$\mathbf{-0.010}$
4 Hull–White Δ_{HW}	0.464	—	$\mathbf{-0.013}$	$\mathbf{-0.006}$
BS Δ–vega	0.246	0.511	-0.049	$+0.036$
Heston Δ–vega	-0.050	1.014	$\mathbf{-0.002}$	$\mathbf{-0.003}$

Daily financing and dividend carry are omitted, as in the notebook's worked example.

Observed SPY and AAPL options

SPY: November 5, 2026 $760 call

SPY261105C00760000. Invert the option close to obtain Black–Scholes implied volatility and delta, taking $r = q = 0$ and calendar days to expiry divided by 365.

2026 date	SPY close	Call close	Volume	IV (%)	Δ^{BS}	h
Oct. 30	$755.72	$4.77	4,626	17.27	0.4036	40
Nov. 2	$752.10	$2.89	5,320	22.07	0.3043	30
Nov. 3	$754.35	$2.89	5,910	23.42	0.3366	34
Nov. 4	$766.81	$8.38	3,989	25.77	0.7480	75

Here $h = \text{round}(100\Delta^{BS})$ shares borrowed and sold short against one long call.

AAPL: November 6, 2026 $345 call

`AAPL261106C00345000`. We use the same four dates and $r = q = 0$ convention as for SPY.

2026 date	AAPL close	Call close	Volume	IV (%)	Δ^{BS}	h
Oct. 30	$344.63	$8.25	3,478	44.28	0.5052	51
Nov. 2	$338.32	$5.25	3,930	57.19	0.3834	38
Nov. 3	$331.90	$0.37	9,524	31.08	0.0869	9
Nov. 4	$349.51	$9.75	6,735	70.94	0.6078	61

Daily hedge errors with borrowed shares

For a long call and h_t borrowed shares sold short,

$$E^{\text{gross}} = 100\Delta C - h_t \Delta S, \qquad E^{\text{net}} = E^{\text{gross}} - h_t D_{t,t+1} - b\, h_t S_t \frac{d}{365}.$$

Asset	Interval (2026)	h_t	$100\Delta C$	$-h_t\Delta S$	Gross	Loan	Net
SPY	Oct. 30–Nov. 2	40	−188.00	+144.80	−43.20	2.48	−45.68
SPY	Nov. 2–3	30	0.00	−67.50	−67.50	0.62	−68.12
SPY	Nov. 3–4	34	+549.00	−423.64	+125.36	0.70	+124.66
AAPL	Oct. 30–Nov. 2	51	−300.00	+321.81	+21.81	1.44	+20.37
AAPL	Nov. 2–3	38	−488.00	+243.96	−244.04	0.35	−244.39
AAPL	Nov. 3–4	9	+938.00	−158.49	+779.51	0.08	+779.43

Values are dollars per standard 100-share contract.

Which call was more closely hedged?

Compare gross residuals within each contract before comparing the assets:

$$R = 1 - \frac{\sum_t (E^{\text{gross}}_{t,t+1})^2}{\sum_t (100\Delta C_t)^2}, \qquad P = \frac{\sqrt{\frac{1}{3}\sum_t (E^{\text{gross}}_{t,t+1})^2}}{100\, C_{\text{Oct 30}}}.$$

Call	Initial premium	Unhedged RMSE	Gross RMSE	R	P
SPY	\$477	\$335.03	\$85.90	93.4%	18.0%
AAPL	\$825	\$634.56	\$471.76	44.7%	57.2%

- The comparison excludes the assumed borrowing fees and does not establish which stock is generally easier to hedge.
- SPY has the smaller normalized hedge error over these three intervals.

More tickers + More techniques

Recalling the terms

Situation : Hold one long option; short h shares and, for the last rule, n units of a longer-dated near-the-money call H.

Hedge	h	n
Black–Scholes	Δ_{BS}	0
Heston	$\partial_S C$	0
Heston minimum variance	$\partial_S C + \frac{\rho\xi}{S}\,\partial_v C$	0
Hull–White	$\Delta_{BS} + \frac{\mathcal{V}_{BS}}{S\sqrt{\tau}}(a + bd + cd^2)$	0
Heston delta–vega	$\partial_S C - n\,\partial_S H$	$\partial_v C / \partial_v H$

Five underlyings: data, calibration, score

Calibration. Refit Heston on each date by vega-weighted least squares. A log-scale prior moderates changes in (κ, θ, ξ). A new run should check convergence and fit error before accepting a date.

Error. Hedge on date t and mark the position at the next trading date:

$$e = \delta C - h\,\delta S - n\,\delta H - r\,\delta t\,(C - hS - nH) - h\,D.$$

Score. Earlier dates fit Hull–White; later dates measure $G_m = 1 - \sum e_m^2 / \sum e_{BS}^2$. The saved notebook uses method-specific option-day rows. A head-to-head comparison requires the same rows for every available hedge. Date-resampled intervals describe uncertainty on a short realized path.

Data audit and calibrated parameters

	Dates	Quotes/date	Date pairs	Option-days	2nd option	Test dates
SPY	18	12	17	322	0%	9
QQQ	9	8	8	147	0%	4
AAPL	54	19	53	971	66%	27
MSFT	17	10	16	296	74%	8
TSLA	22	13	21	340	88%	11

Heston parameters, median over dates:

	ρ	ξ	κ	θ	Feller fails
SPY	-0.605	1.568	6.039	0.046	100.0%
QQQ	-0.304	2.059	5.217	0.078	100.0%
AAPL	-0.380	0.649	3.947	0.094	31.5%
MSFT	-0.255	0.629	3.884	0.120	35.3%
TSLA	$+0.084$	2.010	3.875	0.268	77.3%

2nd option: fraction of option-day rows with a usable second call; intervals require eight test dates per method.

Test-window hedge errors

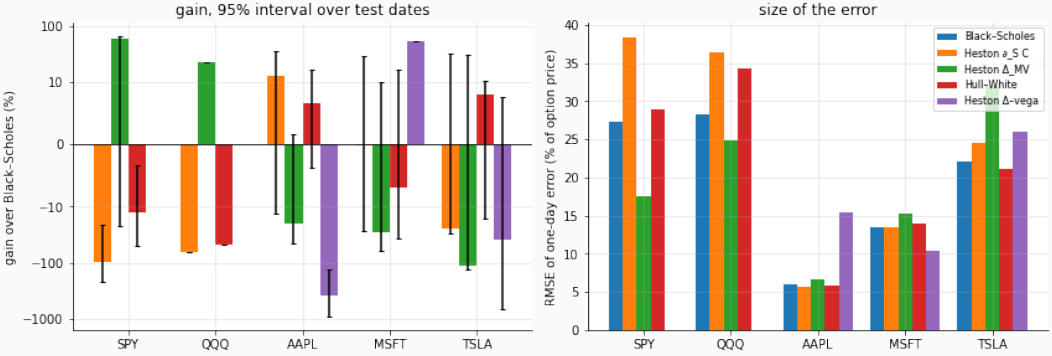


Figure 1: Gain and size of the error.

Share-hedge comparison by underlying

Saved gain over Black–Scholes (%) from the attached notebook. Each method is compared with Black–Scholes on its own option-day rows.

	Heston $\partial_S C$	Heston Δ_{MV}	Hull–White	Heston Δ–vega	Test dates
SPY	-98.2	+58.4	-12.4	–	9
QQQ	-65.2	+22.4	-47.3	–	4
AAPL	+13.0	-20.4	+6.6	-374.1	27
MSFT	-0.1	-28.9	-6.9	+52.5	8
TSLA	-24.3	-111.1	+8.0	-39.5	11

No common-sample share-hedge winner can be identified from the saved outputs: methods may use different option-day rows. QQQ has only four test dates.

Conclusions

Conclusions

- For the prescribed Heston shocks, the minimum-variance and fitted Hull–White share positions have errors of about one cent per option. The plain Heston spot delta has larger errors because it holds variance fixed.
- Adding a one-year Heston hedge call leaves errors of $-\$0.002$ and $-\$0.003$ for the two synthetic shocks. This is a model-based scenario, not a held-out backtest.
- In the observed three-interval sample, the SPY call's gross squared-error reduction is 93.4%; the AAPL call's is 44.7%. AAPL's large move across the strike contributes to the difference.
- One contract per ticker and date identifies an implied-volatility Black–Scholes delta, but not an empirical Heston or Hull–White hedge ratio.
- The saved five-underlying outputs are a dated-sample illustration with an unknown run date; the live panel was not reproduced here.

Limitations and next steps

Limits of this study

- SPY–AAPL: three close-to-close changes per call; SPY expires one day before AAPL
- Calls and puts are American; the models are European
- Last trades may not coincide with stock closes; no bid–ask spreads
- κ is poorly identified even with the prior

Next steps

- Check with American formulas.
- Save the chain every day to lengthen the panel and tighten the intervals
- Include spreads, financing, dividends and broker stock-loan terms
- Check whether the Hull–White coefficients are stable across training windows

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- Yahoo finance for Stock data and AI agents like ChatGPT and Claude for helping me with the code and proofreading the documents.